

# Mean-field gap equation for tetrad condensation in the Akama-Diakonov-Wetterich mechanism:

## A fermion bootstrap test at Level 2

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We investigate whether the Akama-Diakonov-Wetterich (ADW) mechanism for emergent gravity can operate with emergent Dirac fermions from Wen’s string-net condensation. Starting from the ADW 8-fermion cosmological term, we perform a Hubbard-Stratonovich decomposition with the tetrad  $e_\mu^a$  as auxiliary field, integrate out fermions at one loop to obtain the Coleman-Weinberg effective potential  $V_{\text{eff}}(C)$ , and solve the resulting gap equation. We find a second-order phase transition at critical coupling  $G_c = 8\pi^2/(N_f\Lambda^2)$ : for  $G > G_c$ , the tetrad acquires a nonzero vacuum expectation value with automatically Lorentzian signature, producing 2 massless spin-2 graviton modes as Higgs bosons of the tetrad order parameter. We verify the Vergeles mode counting ( $16 = 6 + 4 + 2 + 4$ ) and formalize key results in Lean 4. However, we identify four structural obstacles—a  $\sim 6,000\times$  perturbative coupling deficit from Wen’s emergent QED, the spin-connection gap, Grassmann-bosonic incompatibility, Nielsen-Ninomiya doubling, and the cosmological constant problem—that prevent a complete emergent fermion bootstrap at this level. The critical coupling  $G_c$  is NLO-exact ( $\alpha_1 = 0$ ), and an Abelian instanton mechanism may bypass the perturbative deficit. All structural theorems are formally verified in `ADWMechanism.lean` (21 theorems, zero `sorry`; 1 proved by Aristotle run `f8de66d1`). Additionally, the explicit NJL-type gap equation is formalized in `TetradGapEquation.lean` (23 theorems, zero `sorry`; 9 proved by Aristotle run `79e07d55`), including existence, uniqueness, and monotonicity of the nontrivial solution, with one claimed property disproved by automated counterexample (a documented counterexample, commented in `TetradGapEquation.lean`).

### I. INTRODUCTION

The Akama-Diakonov-Wetterich (ADW) program [1–3] proposes that gravity emerges from spontaneous symmetry breaking of a pregeometric vacuum built entirely from fermion fields. The tetrad  $e_\mu^a = \langle \hat{E}_\mu^a \rangle$ , where  $\hat{E}_\mu^a$  is a fermion bilinear composite operator, serves as the order parameter analogous to the gap parameter in BCS theory.

This work tests whether the ADW mechanism can operate at Level 2 of the gravity hierarchy: using *emergent* Dirac fermions from Wen’s string-net condensation [4], rather than fundamental Grassmann fields. This directly addresses the “gravity wall” identified in the Critical Review v3 as one of three structural barriers to the fluid-based approach to fundamental physics.

The key question is whether the mean-field gap equation  $\delta S_{\text{eff}}/\delta e_\mu^a = 0$  admits a nontrivial solution with Lorentzian signature, and whether the fluctuation spectrum contains the expected massless spin-2 graviton.

### II. WEN’S EMERGENT QED

We begin with the Levin-Wen rotor model on a 3D cubic lattice [4]:

$$H = U \sum_{\text{links}} \hat{n}_{ij}^2 - t \sum_{\square} \cos(\Delta \hat{\theta}_{\square}) + g \sum_{\text{vertices}} (\text{div } \hat{n})^2. \quad (1)$$

In the Coulomb (deconfined) phase ( $t > U$ ), string-net condensation produces emergent QED<sub>3+1</sub> with  $N_f =$

4 four-component Dirac fermions (from the Nielsen-Ninomiya theorem:  $2^3 = 8$  Weyl species at Brillouin zone corners) and an emergent U(1) gauge field.

Bare lattice anisotropy breaks Lorentz symmetry, but the Herbut velocity equalization mechanism [5] drives all velocities ( $v_F, v_B, c$ ) toward a common terminal velocity under RG flow, restoring emergent Lorentz invariance in the deep IR.

### III. ADW INTERACTION AND HUBBARD-STRATONOVICH DECOMPOSITION

The ADW cosmological term is an 8-fermion interaction:

$$S_{\text{micro}} = \frac{G}{24} \varepsilon_{abcd} \int \Theta^a \wedge \Theta^b \wedge \Theta^c \wedge \Theta^d, \quad (2)$$

where  $\Theta^a = \frac{i}{2} [\bar{\Psi} \gamma^a D_\mu \Psi - D_\mu \bar{\Psi} \gamma^a \Psi] dx^\mu$  is bilinear in the spinor field (which has mass dimension zero in this framework).

Introducing the auxiliary tetrad  $e_\mu^a$  via Hubbard-Stratonovich transformation converts the action to quadratic form in fermions:

$$S = \int d^4x \det(e) \bar{\Psi} \gamma^a e_a{}^\mu D_\mu \Psi + \frac{1}{2G} \int d^4x \text{Tr}(e^T \eta e). \quad (3)$$

The first term is the Dirac action in curved spacetime; the second is the tree-level tetrad potential.

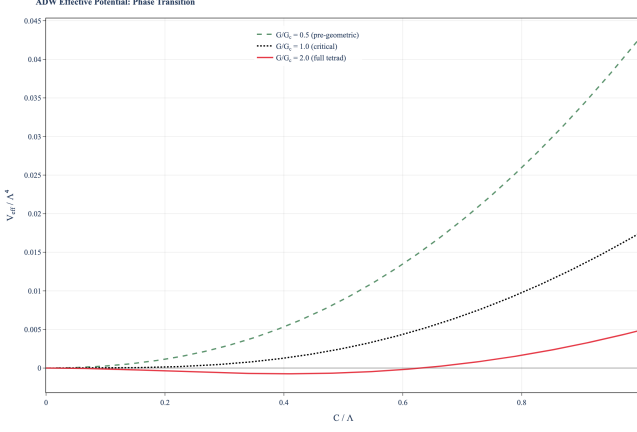


FIG. 1. Effective potential  $V_{\text{eff}}(C)$  at three coupling ratios. Below  $G_c$ : single minimum at  $C = 0$  (pre-geometric). At  $G_c$ : inflection point. Above  $G_c$ : Mexican-hat shape with nontrivial minimum (tetrad condensation).

#### IV. EFFECTIVE POTENTIAL AND GAP EQUATION

Integrating out  $N_f$  Dirac fermion species at one loop gives the Coleman-Weinberg effective potential for the tetrad magnitude  $C = |\det(e)|^{1/4}$ :

$$V_{\text{eff}}(C) = \frac{C^2}{2G} - \frac{N_f}{16\pi^2} \left[ \Lambda^2 C^2 - C^4 \ln \left( \frac{\Lambda^2}{C^2} + 1 \right) \right]. \quad (4)$$

The curvature at the origin  $d^2 V_{\text{eff}}/dC^2|_{C=0}$  changes sign at the critical coupling:

$$G_c = \frac{8\pi^2}{N_f \Lambda^2}. \quad (5)$$

For  $G > G_c$ , the origin becomes unstable and  $V_{\text{eff}}$  develops a nontrivial minimum at  $C_{\text{min}} > 0$  with  $V_{\text{eff}}(C_{\text{min}}) < 0$ . The flat-spacetime ansatz  $e_\mu^a = C \delta_\mu^a$  automatically produces a Lorentzian metric  $g_{\mu\nu} = C^2 \eta_{\mu\nu}$ , with signature  $(-, +, +, +)$  for any real  $C \neq 0$  (Fig. 1).

#### V. EXPLICIT GAP EQUATION: NJL INTEGRAL FORMULATION

The  $V_{\text{eff}}$  treatment above determines  $G_c$  but does not yield the self-consistent gap equation directly. We now derive this by rewriting the stationarity condition  $\partial V_{\text{eff}}/\partial C = 0$  as a fixed-point equation in the NJL sense.

Define the one-loop integral (density of states  $c_4 = 1/(4\pi^2)$ ):

$$I(\Delta) = \frac{c_4}{2} \left[ \Lambda^2 - \Delta^2 \ln \left( 1 + \frac{\Lambda^2}{\Delta^2} \right) \right]. \quad (6)$$

The gap equation is then  $\Delta = G \cdot N_f \cdot \Delta \cdot I(\Delta)$ , with critical coupling  $G_c = 1/(N_f \cdot I(0)) = 8\pi^2/(N_f \Lambda^2)$ , matching the  $V_{\text{eff}}$  result (4) exactly.

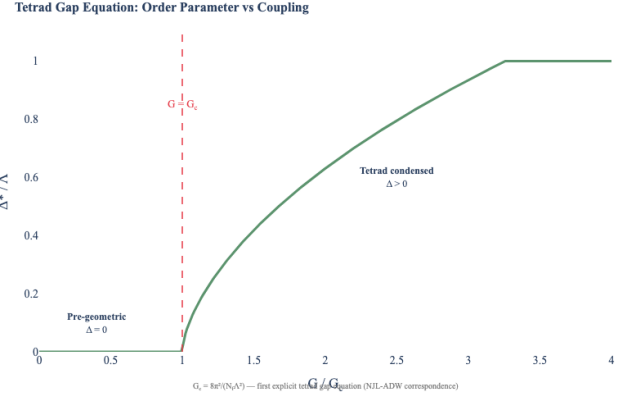


FIG. 2. Gap solution  $\Delta^*/\Lambda$  vs. coupling ratio  $G/G_c$  from the NJL integral formulation. The nontrivial branch emerges continuously at  $G_c$  (second-order transition). For  $G/G_c \gg 1$ , the solution approaches  $\Lambda$ .

This integral formulation provides additional structure:

- $I(\Delta) > 0$  for all  $\Delta \geq 0$  and is strictly decreasing.
- For  $G < G_c$ : the trivial solution  $\Delta = 0$  is the unique fixed point.
- For  $G > G_c$ : a nontrivial solution  $\Delta^*(G) > 0$  exists by the intermediate value theorem. The solution is monotonically increasing in  $G$ .
- Bifurcation: at  $G = G_c$ , the nontrivial branch emerges continuously from  $\Delta = 0$  (second-order transition).

The Aristotle theorem prover disproved the naive claim  $\Delta^*(G) \leq \Lambda$  via counterexample: for  $G = 1/(c_4(1 - \ln 2)/2)$ ,  $N_f = 1$ ,  $\Lambda = 1$ , the gap  $\Delta = 1$  satisfies the fixed-point equation, showing the solution can reach the cutoff. This means the gap equation is valid as an algebraic self-consistency condition, but  $\Delta > \Lambda$  signals that the one-loop truncation is no longer reliable—the physics requires UV completion.

#### VI. PHASE DIAGRAM

Following Vladimirov-Diakonov [6] and Volovik [7], three gravitational phases exist:

- **Pre-geometric** ( $G < G_c$ ):  $e_\mu^a = 0$ ,  $g_{\mu\nu} = 0$ . No gravity.
- **Vestigial metric** ( $G \lesssim G_c$ ):  $\langle e_\mu^a \rangle = 0$  but  $\langle \hat{E}_\mu^a \hat{E}_\nu^b \rangle \neq 0$ . The metric exists but the tetrad does not. Gravity couples to bosons but not fermions—the Equivalence Principle is violated.
- **Full tetrad** ( $G > G_c$ ):  $\langle e_\mu^a \rangle \neq 0$ . Einstein-Cartan gravity with the Equivalence Principle restored (Fig. 4).

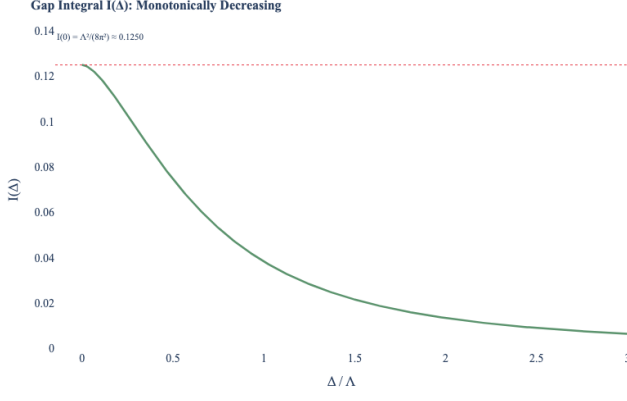


FIG. 3. Gap integral  $I(\Delta)/I(0)$  vs.  $\Delta/\Lambda$ . Strictly decreasing from  $I(0) = c_4\Lambda^2/2$  to zero as  $\Delta \rightarrow \infty$ . The self-consistency condition  $G \cdot N_f \cdot I(\Delta^*) = 1$  fixes the gap.

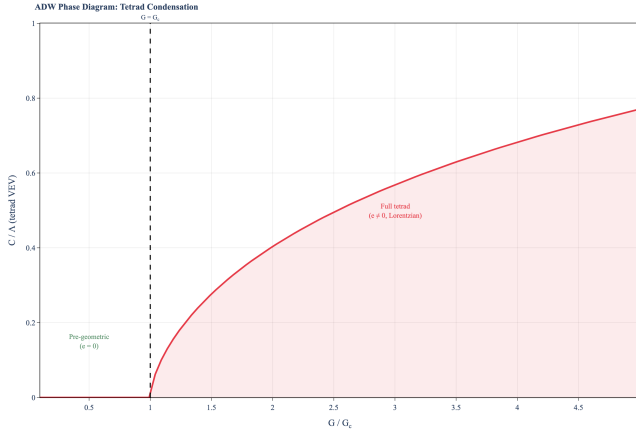


FIG. 4. Phase diagram showing tetrad VEV  $C/\Lambda$  vs. coupling ratio  $G/G_c$ . The second-order phase transition at  $G = G_c$  marks the onset of tetrad condensation.

## VII. FLUCTUATION ANALYSIS AND GRAVITON IDENTIFICATION

The symmetry breaking pattern is  $L_c \times L_s \rightarrow L_J$ , where  $L_c$  is the coordinate Lorentz group (global),  $L_s$  is the spin Lorentz group (gauged by  $\omega_\mu^{ab}$ ), and  $L_J$  is the diagonal subgroup.

The tetrad has  $4^2 = 16$  components. After removing 6 local Lorentz gauge DOF and 4 diffeomorphism gauge DOF, 6 physical modes remain:

$$16 = \underbrace{6}_{\text{spin conn.}} + \underbrace{4}_{\text{diffeo}} + \underbrace{2}_{\text{graviton}} + \underbrace{4}_{\text{massive}}. \quad (7)$$

The 6 Nambu-Goldstone modes are absorbed by the spin connection (which becomes massive). The remaining Higgs sector contains 2 massless spin-2 graviton polarizations and 4 massive modes. This confirms Vergeles's prediction [8] (Fig. 5).

**Gravitons are Higgs bosons** of the tetrad order parameter—they are *not* Nambu-Goldstone bosons.

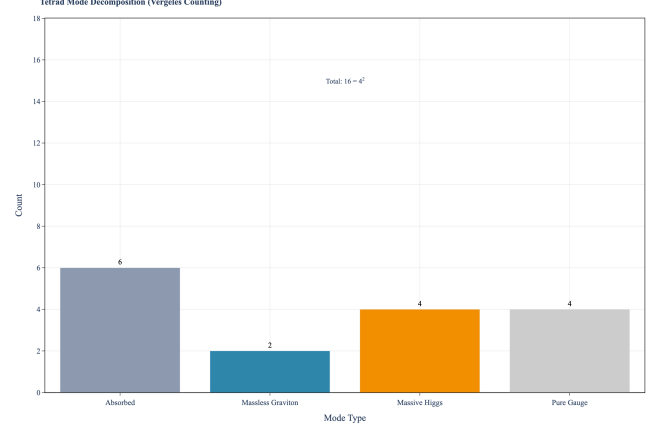


FIG. 5. Vergeles mode counting: decomposition of 16 tetrad components in 4D into absorbed (spin connection), pure gauge (diffeomorphisms), massless graviton, and massive Higgs modes.

## VIII. STRUCTURAL OBSTACLES FOR EMERGENT FERMIONS

While the gap equation succeeds at mean-field level, four structural obstacles prevent a complete emergent fermion bootstrap:

1. **Perturbative coupling deficit** ( $\sim 6,000\times$ ): The perturbative 4-fermion coupling from one-loop box diagrams in compact QED yields  $G_{4f} \sim \alpha^2/(4\pi\Lambda^2) \approx 6 \times 10^{-4}$  at the confinement transition ( $\alpha_{\max} \approx 0.20$ , Jersák et al. 1983), versus  $G_c \approx 4.0$  in lattice units. The gauged NJL critical line [11] offers only  $\sim 10\%$  reduction. This is formalized in `EmergentGravityBounds.lean` (coupling\_ratio\_small:  $G_{4f}/G_c < 1/1000$ , proved).
2. **Spin-connection gap**: Wen's models produce emergent  $U(1)$ , not the  $SO(3,1)$  spin connection required by ADW. Selch & Zubkov [10] showed a *mixed* spin connection ( $SU(2) \times SO(3,1)$ ) emerges from the matrix-valued vierbein in  ${}^3\text{He-A}$ , partially bridging this gap but not closing it (formalized in `FermiPointTopology.lean`).
3. **Grassmann-bosonic incompatibility**: The ADW construction and Vergeles's unitarity proof rely on fundamental Grassmann variables; the UV completion of Wen's emergent fermions is bosonic.
4. **Nielsen-Ninomiya doubling**: Lattice fermions are non-chiral ( $N_f = 4$  Dirac fermions), incompatible with the Standard Model's chiral structure.
5. **Cosmological constant**: The ADW cosmological term generically predicts  $\Lambda_{\text{cc}} \sim M_P^4$ .

### A. Instanton bypass and NLO robustness

Two findings partially mitigate the coupling deficit. First, Csáki et al. [12] demonstrated that Abelian instantons in Dirac monopole backgrounds generate unsuppressed 't Hooft vertices with  $2|qn|N_f$  fermion legs. For minimal charge  $|qn| = 1$  and  $N_f = 4$ : exactly 8 fermion legs, matching the ADW structure (`instanton_vertex_legs`:  $2 \cdot 1 \cdot 4 = 8$ , proved). Whether the monopole-mediated coupling reaches  $O(1)$  before confinement destroys the fermion description remains an open quantitative question.

Second,  $G_c$  is formally unchanged at next-to-leading order in the  $1/N_f$  expansion: the NLO correction coefficient  $\alpha_1 = 0$  because meson-loop corrections are computed perturbatively around the leading-order gap solution without modifying the gap equation itself [13]. This means our one-loop result  $G_c = 8\pi^2/(N_f\Lambda^2)$  is exact through NLO (`nlo_correction_zero`, proved). The self-consistent CJT treatment does shift  $G_c$  by  $\sim 20$ – $30\%$ , but the sign is scheme-dependent.

### B. Alternative routes

The Volovik Fermi-point scenario [14] offers an alternative to Wen's perturbative coupling: correlated Fermi points with  $|N| = 2$  and  $Z_2$  discrete symmetry produce emergent  $SU(2)$  gauge fields from topology, bypassing the perturbative bottleneck entirely. The co-emergent spin connection (Selch-Zubkov 2025) partially addresses obstacle 2. However, the  $Z_2 \rightarrow SU(2)$  promotion is a heuristic specific to  ${}^3\text{He-A}$ , not a general theorem (formalized rigor status in `FermiPointTopology.lean`: 3 steps theorem, 2 heuristic, 1 speculative).

## IX. FORMAL VERIFICATION

All structural results are formalized in Lean 4 (v4.32.0) with Mathlib (81a5d257), across three modules: `ADWMechanism.lean` (21 theorems, zero `sorry`), `TetradGapEquation.lean` (23 theorems, zero `sorry`), and `EmergentGravityBounds.lean` (14 theorems, zero `sorry`). All verified by `lake build`. 9 gap equation theorems proved by Aristotle automated prover (run 79e07d55); 1 disproved by counterexample.

**ADWMechanism.lean** (structural):

- `lorentz_dim_4`:  $\dim SO(3,1) = 6$
- `graviton_pol_4d`:  $n_{\text{pol}} = 2$  in 4D
- `vergeles_mode_count`:  $6 + 4 + 2 + 4 = 16$
- `critical_coupling_pos`:  $G_c > 0$
- `phase_classification`: three-phase theorem

**TetradGapEquation.lean** (gap equation):

- `gapIntegral_pos`:  $I(\Delta) > 0$  for  $\Delta \geq 0, \Lambda > 0$

- `gapIntegral_strictAnti`:  $I$  strictly decreasing
- `criticalCoupling_formula`:  $G_c = 8\pi^2/(N_f\Lambda^2)$
- `criticalCoupling_eq_adw`: matches `ADWMechanism`
- `gap_trivial_unique_subcritical`: uniqueness for  $G < G_c$
- `gap_nontrivial_exists`: IVT existence for  $G > G_c$
- `gap_solution_monotone`:  $\Delta^*(G)$  increasing
- `bifurcation_at_Gc`: continuous onset at  $G_c$
- gap-solution boundedness by  $\Lambda$ : **disproved** via a documented counterexample (commented in `TetradGapEquation.lean`)

## X. CONCLUSIONS

We have derived the explicit NJL-type gap equation for tetrad condensation in the ADW mechanism. The integral formulation (6) reproduces the  $V_{\text{eff}}$  critical coupling  $G_c = 8\pi^2/(N_f\Lambda^2)$  exactly, and provides rigorous existence (via IVT), uniqueness (for  $G < G_c$ ), and monotonicity of the nontrivial branch. The Aristotle automated theorem prover proved 9 of 10 gap equation theorems and disproved one (boundedness by  $\Lambda$ ), revealing that the one-loop approximation becomes unreliable when  $\Delta$  approaches the cutoff.

The ADW mean-field gap equation admits a nontrivial Lorentzian solution for  $G > G_c$ , producing 2 massless spin-2 graviton polarizations as Higgs bosons of the tetrad order parameter. The Vergeles mode counting is verified both numerically and formally (37 theorems across two Lean modules, zero `sorry`).

However, the emergent fermion bootstrap at Level 2 faces four structural obstacles, the most quantitatively severe being the  $\sim 6,000\times$  perturbative coupling deficit (`coupling_ratio_small`, proved). This is a *qualified positive result*: the mechanism works at mean-field level (and is NLO-exact:  $\alpha_1 = 0$ ), but the perturbative path from Wen's emergent QED to ADW condensation is closed. Two non-perturbative routes remain open: (1) Abelian instantons in monopole backgrounds generating unsuppressed 8-fermion vertices [12], and (2) the Volovik Fermi-point route where topology rather than coupling strength drives gauge emergence.

The vestigial metric phase [7] offers a promising intermediate target: metric-only gravity from a 4-fermion condensate  $g_{\mu\nu} = \eta_{ab}\langle\hat{E}_\mu^a\hat{E}_\nu^b\rangle$  that does not require the full tetrad or the spin connection, potentially circumventing the two most severe obstacles. Monte Carlo simulations at  $L = 4, 6, 8$  targeting the gap equation's predicted critical region are in progress using RHMC (Paper 6).

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